

APEX RETAIL

AI Store Intelligence & Re-ID Shopper Journey Analytics

Built for the Purple Tech Challenge 2026 | Round 2 Individual Submission

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Live Demo: <https://shahsuraj0204.github.io/store-intelligence/>

Core Pillars: Edge Computer Vision • Cross-Camera Re-ID • POS Basket Analytics • AI Voice Copilot

Retail Visibility Bottlenecks

Traditional brick-and-mortar stores operate in a visual blindspot regarding customer behavior.

- **Blind Dwell Times:** High traffic aisles are seen, but shelf-level drop-offs and product affinities remain completely unknown.
- **Checkout Queue Abandonment:** Checkout queues spike rapidly without predictive notifications, resulting in lost sales and frustrated shoppers.
- **Static Store Layouts:** Product placements are determined by visual intuition rather than actual empirical correlation with customer paths.

Edge-to-Cloud System Flow

A lightweight, database-backed processing pipeline running efficiently on commodity hardware.

- **Edge CV Pipeline:** YOLOv8 object detection + ByteTrack frame-skipped inferences running on standard CPU feeds.
- **FastAPI REST & WebSockets:** Dynamic ingestion endpoints storage with SQLite, broadcasting updates asynchronously.
- **Interactive Web Dashboard:** Modern glassmorphism dark-themed dashboard showing live streams, conversion funnels, occupancy, and anomalies.

Spatial Re-ID Journey Timeline

Tracing the exact movements of individual customer sessions across store cameras.

- **Stitched Pathways:** Resolves localized camera track IDs to a single global Visitor ID across non-overlapping feeds (Entrance -> Skincare -> Makeup -> Billing -> Exit).
- **Behavioral Telemetry:** Extracts precise dwell durations at shelves, billing queue wait times, and movement velocities.
- **Traffic Segmentation:** Identifies and filters employee tracks dynamically to ensure layout analysis is based purely on shopper behavior.

POS Basket Correlation A/B

Linking physical shopper paths directly with final transaction details.

- **POS Transaction Ingestion:** Analyzes transaction CSV files to extract buy-together product affinities (association rule mining).
- **Physical Flow Maps:** Tracks actual customer walk-through rates between related display racks (e.g., Haircare to Fragrance).
- **Placement Efficiency Index (PEI):** Automatically flags layout bottlenecks (e.g. shelves with high buy-together affinity but low physical shopper transitions) to guide physical relocations.

Interactive Voice & Sound Copilot

An eyes-free, voice-assistant desk for multi-tasking store managers.

- **Web Audio Synthesizer:** Synthesizes clean sound chime pings and warning audio alerts directly inside the web browser.
- **TTS Notifications:** Automatically reads operational recommendations out loud (e.g., checkout queue build-ups) so managers can allocate staff instantly.
- **Auditory Alarms:** Plays real-time sirens when security breaches (unauthorized restricted office entries) are detected by the computer vision feed.